

IGME 382: Project Proposal

1. Topic & Spatial Question

Topic:

Evaluate neighborhoods for home buying in the Seattle area based on details like proximity to amenities such as restaurants, parks, trails, grocery stores, and other amenities.

Spatial Question:

Which residential parcels in Seattle best match a set of neighborhood home buying requirements like walkability to restaurants, access to parks, quiet streets, proximity to walking trails. How does suitability vary across the city?

2. Data Sources & Planned Modifications

Dataset 1: King County Parcels / parcel_area

- What it represents:
A polygon dataset of every property parcel in King County, including land boundaries for residential, commercial, industrial, parks, government properties, and vacant land. This dataset will be used for identifying houses and show statistics for each property
- Link:
<https://gis-kingcounty.opendata.arcgis.com/maps/8058a0c540434dadbe3ea0ade6565143>

Dataset 2: OpenStreetMap Amenities for North America (Living Atlas)

- What it represents:
A point layer derived from OpenStreetMap containing thousands of “amenity” features categorized by type, including restaurants, cafés, bars, grocery stores, schools, bike shops, and other useful walkable destinations. This dataset will be used for calculating suitability of parcels based on their proximity to these amenities.
- Link:
<https://www.arcgis.com/home/item.html?id=d3e9c2f8cdcd41f38e64b54c3a9a1c74>

Dataset 3: Parks in King County / park_area

- What it represents:
A polygon dataset representing parks, recreation spaces, greenspaces, and public open areas across King County. This will be used for calculating suitability based on their proximity to these parks.
- Link:
<https://gis-kingcounty.opendata.arcgis.com/maps/a0c94c33228146c5ad95a1dff3b6963d>

Planned Modification in ArcGIS Pro

I will be using ArcGIS Pro mainly for preparing the data beforehand. This will include modifying the parcel dataset before using it for analysis. First I will clip the King County parcel layer to a desired area (Seattle city boundary) to limit the study area. Next, I will filter the dataset to only include residential parcels using the land use or property type fields. I will then export this as a new dataset "Seattle_Residential_Parcels" which will serve as the foundation for my analysis. Doing this modification will make sure that the data only includes relevant residential properties.

3. Additional Geospatial Tool (Beyond ArcGIS Pro)

Tool:

Python/GeoPandas

How I Will Use It:

I will use Python to automate the suitability analysis for each residential parcel. First the script will ask for a base dataset of the parcels you want to evaluate like "Seattle_Residential_Parcels" along with the parks and amenities layers. After this the user will enter their desired amenities from a selectable list, as well as how far they are willing to walk or drive to the selected amenities. These variables will then be used to compute in real time buffers around the selected amenities such as parks and restaurants. Then for each parcel it will calculate how many amenities of each selected type are in walking distance and generate a score based on user defined priorities. The script will output a scored layer table including how many of each amenity was in walking distance to be loaded back into ArcGIS Pro for visualization as a heat map.

4. Expected Final Product

My plan is to create a by parcel heat map showing which residential properties in Seattle best match a set of home buying requirements. The final output will include an ArcGIS Online web map where users can visually explore the heat map, click on statistics like how many restaurants are within walking distance, distance to nearest grocery stores, and proximity to parks. This will help show how different areas rank in terms of livability and clearly show using spatial data neighborhood suitability across the city.

5. Inspiration

This project is inspired by my sister's recent home buying search in Seattle and the criteria she shared: walkability to restaurants, access to parks and trails, and proximity to daily amenities. Her notes directly shaped the suitability factors I'm analyzing.

Neighborhood needs:

- 5-10 min walk to some restaurants
- 5-10 min drive to a wide variety of restaurants/grocery store
- Close to biking trails
- Quieter street - next to main streets, but not on one
- Big trees, veggies growing in the neighborhood
- Room to grow the backyard into an outdoor kitchen/living room with a tv and a firepit situation

Friday 17:11

I think proximity to a park or greenspace would be another fun thing to filter for, especially for families with kids